COURTNEY M. CURRIER, Ph.D.
Curriculum Vitae

Department of Earth System Science, Stanford University, Stanford, CA 94305
ccurrier@stanford.edu | https://courtcurrier.github.io

RESEARCH INTERESTS

I am an experimental plant ecologist focused on plant–soil–microbial relationships under global change. I use a combination of field experiments, isotope tracers, and large-scale spatial analyses to link ecological processes across scales and predict ecosystem responses to disturbance and climate change. Ultimately, my work aims to improve predictions of ecosystem nutrient and carbon dynamics to better inform the role of terrestrial ecosystems, especially drylands, in natural climate solutions.

ACADEMIC QUALIFICATIONS

2018–2023	Ph.D. Environmental Life Sciences, Arizona State University, Tempe, AZ
2013–2015	M.Sc. Biology, Arizona State University, Tempe, AZ
2009–2013	B.Sc. Environmental Sciences (Honors Research), University of Notre Dame, Notre Dame, IN

PROFESSIONAL EXPERIENCE

2026–present	Postdoctoral Scholar, Department of Earth System Science, Stanford University
2023–2026	Postdoctoral Research Associate, Department of Plant Sciences, University of Cambridge, United Kingdom
2025	Lecturer, University of Cambridge
2017–2018	Project Manager, Global Drylands Center, School of Sustainability, Arizona State University
2015–2018	Research Specialist and Lab Manager, Sala Laboratory, Arizona State University

FELLOWSHIPS AND GRANTS

($82,290 awarded)

2025	Stanford University C-ShaRP Voucher (co-PI)
2023	National Science Foundation Postdoctoral Fellowship in Biology: Bridging the gap between ecosystem vulnerability to nitrogen loss and opportunity for uptake (rejected, overall rating of 'Very Good')
2022	Presidential Management Fellowship (rejected, semifinalist)
2022	PhD Completion Fellowship, School of Life Sciences, Arizona State University
2022	Graduate College Travel Award, Arizona State University
2020	Student Registration Award, Ecological Society of America Biogeosciences Section
2019–2021	Jornada LTER Graduate Research Fellowship
2019	Student Travel Award, Ecological Society of America Southwest Section

2019	Research Training Initiative Grad Student Support Grant, Arizona State University
2018–2019	Arizona State University Graduate College Fellowship
2018	Excellence in Rangeland Ecology Research Travel Award, Ecological Society of America Rangeland Ecology Section
2018	Arizona State University, Recruitment Fellowship
2017	Travel Grant Award, ASU, Graduate and Professional Student Association
2015	Travel Grant Award, ASU, School of Life Sciences
2015	Travel Grant Award, ASU, Graduate and Professional Student Association
2013	National Science Foundation – Graduate Research Fellowship (rejected, Honorable Mention)
2012	National Science Foundation – Research Experience for Undergraduates Fellowship, Niwot Ridge LTER
2012	Travel Grant Award, University of Notre Dame, College of Science
2012	Travel Grant Award, National Science Foundation – Biology REU

HONORS AND AWARDS

2022	Achievement Rewards for College Scientists (semifinalist)
2021	"Grad Slam" Elevator Pitch 3rd Place Winner, Arizona State University Graduate and Professional Student Association
2020	Outstanding Research Award, Arizona State University Graduate and Professional Student Association
2018–2022	The Graduate College Excellence Award, Arizona State University
2013	Outstanding Environmental Scientist Award, University of Notre Dame
2012–2013	Honors Research Student in Biological Sciences, University of Notre Dame
2011–2012	Dean's List, University of Notre Dame, College of Science
2011	Clare Boothe Luce Undergraduate Research Award, University of Notre Dame College of Science Summer Research Fellow

PEER-REVIEWED PUBLICATIONS

(H index = 9 | Google Scholar)

Currier, C.M., Parrinello, C., Dick, C., Coetsee, C., Czimczik, C., Modimola, K.S., Moreland, K., Wigley, B., dos Santos Winston, G., and A.F.A. Pellegrini. 2026. Soil stabilization predicts decoupling between plant productivity and ecosystem carbon storage. In preparation for submission to *Nature Geoscience*.

Currier, C.M., Reichmann, L.G., and O.E. Sala. 2026. Unresponsive drylands to nitrogen availability: Access to alternative sources? In preparation for submission to *Journal of Geophysical Research: Biogeosciences*.

Currier, C.M., Parrinello, C., Dick, C., Augustine, D.J., Blair, J.M., Boughton, E.H., Coetsee, C., David, A., Derner, J.D., Hobbie, S.E., Isbell, F., Modimola, K.S., Reich, P.B., Wigley, B. Worm, K., Zeglin, L.H., and A.F.A. Pellegrini. 2026. Belowground productivity in grasslands and savannas depends on shifting fire and herbivory regimes. *Global Change Biology*, accepted.

Powell, R., …, **Currier C.M**., …, et al. 2026. Constraints and opportunities for improved rangeland grazing. *Science*, 392(6803): 1161-1166. DOI

Currier, C.M., Reichmann, L.G., and O.E. Sala. 2026. Acclimation of the nitrogen cycle to changes in precipitation. *Ecosystems* 29(25). DOI

Biancari, L., ..., **Currier, C.M.**, ..., et al. 2024.. Drivers of woody dominance across global drylands. *Science Advances* 10,eadn6007. DOI

Díaz-Martínez, P., ..., **Currier, C.M.**, ..., et al. 2024. Vulnerability of mineral-associated soil organic carbon to climate across global drylands. *Nature Climate Change* 14: 976–982. DOI

Maestre, F., …**Currier, C.M.**, …, et. al. 2022. Grazing and ecosystem service delivery in global drylands. *Science* 378: 915-920. DOI

Currier, C.M. and O.E. Sala. 2022. Precipitation versus temperature as phenology controls in drylands. *Ecology*. DOI

Osborne, B. B., Bestelmeyer, B. T., **Currier, C. M.**, Homyak, P. M., Throop, H. L., Young, K., and Reed, S. C. 2022. The consequences of climate change for dryland biogeochemistry. *New Phytologist*. DOI

Currier, C.M., Makings, E., Anderson, J., Slagel, K., and J. Maranville. 2021. Solanaceae Part Seven: Browallia, Calibrachoa, Capsicum, Jaltomata, and Salpichroa. *Canotia* 17: 46-60. DOI

Currier, C.M., Chaloner, D.T., Ruegg, J., Tiegs, S.D., D'Amore, D., and G.A. Lamberti. 2020. Beyond macronutrient resource subsidies: Pacific salmon (Oncorhynchus spp.) as potential vectors of micronutrients. *Aquatic Sciences* 82:50. DOI

Sala, O.E., Boone, C.G., Turner II, B.L., and **C.M. Currier**. 2019. The sustainability gap and its implications. *Current Opinion in Environmental Sustainability* 39: 39-43. DOI

Delgado-Baquerizo, M., ..., **Currier, C.M.**, ..., et al. 2019. Changes in belowground biodiversity during ecosystem development. *Proceedings of the National Academy of Science* 116(14): 6891-6896. DOI

Currier, C.M., and J.J. Elser. 2017. Beyond monoculture stoichiometry studies: assessing growth, respiration, and feeding responses of three *Daphnia* species to P-enriched, low C:P lake seston. *Inland Waters* 7(3): 348-357. DOI

Rüegg, J., **C.M. Currier**, D.T. Chaloner, S.D. Tiegs, and G.A. Lamberti. 2014. Habitat influences Pacific salmon (Oncorhynchus spp.) tissue decomposition in riparian and stream ecosystems. *Aquatic Sciences* 76(4): 623-632. DOI

SELECTED ORAL PRESENTATIONS

Currier, C.M., Parrinello, C., Dick, C., Augustine, D.J., Blair, J.M., Boughton, E.H., Coetsee, C., David, A., Derner, J.D., Hobbie, S.E., Isbell, F., Mkansi, N., Modimola, K.S., Reich, P.B., Wigley, B. Worm, K., Zeglin, L.H., and A.F.A. Pellegrini. 2025. Fire and aridity controls on soil carbon inputs. Ecological Society of American Annual Meeting. Baltimore, MD, USA. August 2025. (oral presentation)

C.M. Currier. Soil organic carbon dynamics across pyrogenic landscapes. Archbold Biological Station, Microbial Ecology Workshop. August 2025. (invited talk)

C.M. Currier. Answering burning questions in savannas. Emmanuel College, University of Cambridge, College Research Symposium. February 2025. (invited talk)

C.M. Currier. Dryland ecosystems and their plant adaptations to global change. University of the Third Age, Botany Seminar. January 2025. (invited talk)

C.M. Currier. The wonderful world of drylands. Cambridge Botanical Garden, Science on Sunday Series. July 2024. (invited talk)

C.M. Currier and O.E. Sala. Nitrogen plant uptake in a semiarid ecosystem is modulated by water and plant type. Ecological Society of America Annual Meeting. Montreal, Quebec, Canada. August 2022. (oral presentation)

C.M. Currier and O.E. Sala. A temporal perspective of nitrogen cycling in a semiarid grassland under extreme rainfall conditions. Ecological Society of America Annual Meeting. Virtual. August 2021. (oral presentation)

C.M. Currier and O.E. Sala. Re-envisioning plant phenology: From data points to painting. Ecological Society of America Annual Meeting. Virtual. August 2020. (co-organizer of session)

C.M. Currier and O.E. Sala. Extreme precipitation interacts with N cycling in a semi-arid grassland. Ecological Society of America Annual Meeting. Virtual. August 2020. (oral presentation)

TEACHING

University of Cambridge Lecturer and Supervisor
Part II Plant Sciences Module: Global Change

Graduate Teaching Assistant at Arizona State University
Biology 426: Limnology
Biology 320: Fundamentals of Ecology
Biology 281: General Biology I for Majors
Biology 282: General Biology II for Majors
Biology 181: General Biology I
Biology 182: General Biology II
Biology 100: The Living World

Undergraduate Teaching Assistant at Notre Dame
Biology 30312: General Ecology

OTHER PUBLICATIONS AND ART

C.M. Currier. 2025. At the cutting edge. *Emmanuel Review*. Online only

C.M. Currier and T. Feltus. 2023. Finding joy through art, ecology, and bikes. *The Radavist*. Online only.

THIRST 2022: Artists for Humanitarian Aid. Online only. July 2022 (original paintings)

C.M. Currier. 2019. Perspectives from a field ecologist. *Phoenix Transect*. Online only

Dimensions of Science. Onyx Gallery, Phoenix, AZ. February 2019 (original painting)

C.M. Currier and O.E. Sala. 2018. Art provides a new lens for moving through the scientific method. *SciArt Magazine* 33: How can art influence science? Online only

C.M. Currier. 2018. *Life Raft Zine.* Online only

Portraits of Science. MonOrchid Gallery. Phoenix, AZ. May 2017 (original paintings)

TRAINING

2026	Radiocarbon in Earth Systems Course, Max Planck Institute for Biogeochemistry
2024	Wilderness First Aid Responder and CPR
2019	IsoCamp, University of Utah
2018	Explore National Ecological Observatory Network Workshop, Boulder, CO

PROFESSIONAL ASSOCIATIONS

2026–present	American Geophysical Union, member
2024–present	College Research Associate, Emmanuel College, University of Cambridge
2017–present	Ecological Society of America, member
2017–2024	BIODESERT Global Research Group
2017	CLIMA-FUN belowground biodiversity survey

SERVICE

2024–present	Women in Soil Ecology Mentor
2022–present	Reviewer for *New Phytologist*, *Biogeochemistry*, *Global Change Biology, Ecology* journals
2022	Abstract Reviewer for the Ecological Society of America Southwestern Chapter
2020–2022	Grant Reviewer for the Graduate and Professional Student Association, Arizona State University
2019–2023	Vision Kids Art Program Instructor, Chandler, AZ
2019–2021	Jornada Basin LTER Graduate Student Representative
2017–present	Alumni Mentor, Environmental Sciences undergraduate program, Notre Dame
2016–present	Asombro Institute, New Mexico State University, Jornada LTER, volunteer
2013–2014	Arizona State University, School of Life Sciences Graduate Executive Committee, secretary
2013–present	Arizona State University Ask a Biologist, volunteer
2012–2013	Notre Dame Biological Sciences Senior Leadership Committee